

Task-1 Week-1 Project Documentation

1. Introduction

This document summarizes the work completed during Week 1 of the VLSI internship Task-1.

The main areas covered were:

- Digital electronics fundamentals
 - Linux environment setup
 - Verilog HDL programming
 - Testbench development
 - Icarus Verilog simulation
 - GTKWave waveform analysis
 - Basic digital circuit design
 - Git and GitHub version control
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2. Development Environment

Hardware/Operating Environment

- Windows
- Ubuntu Linux through WSL

Software Tools

- Visual Studio Code
 - Ubuntu WSL
 - Icarus Verilog
 - GTKWave
 - Git
 - GitHub
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3. Digital Electronics

The following logic gates were studied and implemented:

- AND
- OR
- XOR
- NOT
- NAND
- NOR
- XNOR

Truth tables and K-map solutions were documented separately.

4. Verilog HDL Designs

The following circuits were implemented using Verilog HDL:

Logic Gates

1. AND Gate
2. OR Gate
3. XOR Gate
4. NOT Gate
5. NAND Gate
6. NOR Gate
7. XNOR Gate

Combinational Circuits

8. 2-to-1 Multiplexer
 9. Half Adder
 10. Full Adder
 11. 4-bit Ripple Carry Adder
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5. Verification Methodology

Each Verilog design was tested using a dedicated testbench.

The verification flow was:

Verilog RTL ↓ Testbench ↓ Icarus Verilog ↓ Simulation ↓ VCD Waveform
↓ GTKWave ↓ Waveform Verification

A0 — B0 — Cin — Full Adder — S0 | | | C1 ↓ ↓
 A1 — B1 — C1 — Full Adder — S1 | | | C2 ↓ ↓
 A2 — B2 — C2 — Full Adder — S2 | | | C3 ↓ ↓
 A3 — B3 — C3 — Full Adder — S3 | | | Cout

[week-1/documentation/waveforms/](#)

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